BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Data Analysis Report

Project ID: 25-26J-123

Project Title: AI-POWERED DATA ANONYMIZATION FOR THE DEPARTMENT OF CENSUS AND STATISTICS OF SRI LANKA

1. Introduction

1.1. Background:

The Department of Census and Statistics of Sri Lanka collects and manages large volumes of population-level microdata such as household, income, employment, education, and demographic information. These datasets are highly valuable for research, policy planning, economic analysis, and national development decision-making. However, because microdata often contains personal and household-level information, sharing such datasets creates serious privacy risks.

Traditional anonymization techniques such as removing direct identifiers, generalization, suppression, and masking are useful but may not always be sufficient. Even when names, ID numbers, and addresses are removed, individuals can still be re-identified using combinations of quasi-identifiers such as age, gender, district, occupation, religion, education level, or household characteristics.

Therefore, this research focuses on developing an AI-powered data anonymization framework that can support safer data sharing. The system combines multiple privacy-preserving approaches, including NLP-driven identifier detection, differential privacy, permutation-based synthetic data generation, dynamic anonymization technique selection, and machine learning-based re-identification risk evaluation.

1.2. Research Problem:

Government statistical organizations such as the Department of Census and Statistics must share useful microdata for research and policy analysis while protecting the privacy of individuals and households. However, existing manual anonymization processes can be time-consuming, inconsistent, and difficult to scale across different datasets.

A major challenge is that privacy risks do not only come from direct identifiers such as names or ID numbers. Quasi-identifiers such as age, gender, location, occupation, and education level can also lead to re-identification when combined with external or auxiliary datasets. In addition, applying too much anonymization can reduce the usefulness of the dataset, while applying too little anonymization can expose sensitive information.

Therefore, the research problem addressed in this project is:

How can an AI-assisted anonymization framework automatically identify privacy sensitive attributes, generate privacy preserving synthetic data, select suitable anonymization techniques, and evaluate re-identification risk while maintaining the analytical usefulness of census microdata?

1.3. Objectives:

The main objective of this research is to design and develop an AI-powered data anonymization framework that supports secure and privacy-preserving data sharing for census and statistical microdata.

The specific objectives are:

1. **NLP-Driven Identifier and Quasi-Identifier Detection**
   To automatically analyze dataset attributes and classify them as direct identifiers, quasi-identifiers, sensitive attributes, or non-sensitive attributes using statistical and semantic features.
2. **Agentic AI-Based Dynamic Anonymization Technique Selection**
   To recommend suitable anonymization techniques based on dataset characteristics, attribute sensitivity, privacy risk, and expected data utility.
3. **Synthetic Data Generation using Differential Privacy and Permutation Paradigm**
   To generate high-quality synthetic microdata that preserves the statistical properties of the original dataset while reducing the risk of exposing real individual records.
4. **Machine Learning-Based Re-identification Risk Evaluation**
   To evaluate the risk of re-identification in anonymized datasets using machine learning-based attack simulation, similarity analysis, and record-level risk scoring.

It Number	Objective	Objective number
IT22266750	**NLP-Driven Identifier & Quasi-Identifier Detection** 1. Analyze the structure and characteristics of the HIES dataset for identifying different attribute types 2. Classify attributes into Direct Identifiers, Quasi-Identifiers,	1

	Sensitive, and Non-sensitive categories 3. Examine statistical properties such as uniqueness, missing values, and distributions of attributes 4. Identify potential quasi-identifier combinations that can lead to re-identification risk 5. Support feature engineering for the NLP-based classification model 6. Evaluate disclosure risk using basic statistical metrics (uniqueness %, k-anonymity indicators)	
IT22364760	**Agentic AI-Based Dynamic Anonymization Technique Selection** 1. Analyze dataset attributes such as age, gender, location, and income to identify their level of sensitivity and role in privacy risk. 2. Categorize attributes into Direct Identifiers, Quasi-Identifiers, Sensitive, and Non-sensitive to support proper anonymization decisions. 3. Evaluate statistical properties such as uniqueness percentage and distribution patterns to understand which attributes require stronger protection. 4. Select suitable anonymization techniques such as masking, generalization, suppression, or permutation based on attribute type and risk level. 5. Ensure a balance between privacy protection and data utility by applying appropriate techniques without losing important analytical information.	2
IT22129680	**Integrating Differential Privacy With The Permutation Paradigm For Synthetic Data Generation**	3

	1. Analyze real microdata structures to identify attribute types, missing value patterns, variable distributions, correlations, and sensitive data elements that require privacy protection. 2. Generate high-quality synthetic datasets that closely reflect the statistical properties of the original microdata while minimizing the risk of exposing real individual records. 3. Apply privacy-preserving techniques such as differential privacy, permutation-based transformation, and controlled noise addition to protect sensitive and quasi-identifying information during synthetic data generation. 4. Evaluate the generated synthetic data using utility and privacy metrics to ensure that it maintains analytical usefulness while reducing disclosure risks. 5. Ensure alignment between the original dataset characteristics and the synthetic output so that researchers can perform meaningful analysis without accessing confidential real microdata. 6. Improve the effectiveness of secure data sharing by reducing manual anonymization effort and providing structured, privacy-preserving synthetic datasets that support ethical research and policy analysis.	

IT22319074	**4. Re-identification Risk Evaluation:** Evaluate the risk of re-identification in anonymized datasets using machine learning-based attack simulation. Use prediction probabilities generated by trained models to estimate how likely each record can be linked back to an individual. Apply feature-based similarity analysis (such as age difference, gender match, and location similarity) to compare anonymized and auxiliary records. Aggregate pair-level predictions into record-level risk scores using techniques such as maximum and average scoring. Classify records into different risk levels (Critical, High, Medium, Low) using percentile-based thresholds to identify highly vulnerable records. Analyze feature importance to determine which attributes contribute most to re-identification of risk and privacy exposure.	4

2. Data Exploration
   2.1. Data Collection

The main dataset for this research is based on microdata provided by the Department of Census and Statistics of Sri Lanka. The dataset contains demographic, household, income, employment, education, and location-based attributes that are suitable for analyzing privacy risks in census-style records.

In addition to the original dataset, publicly available datasets may be used for experimentation, testing, and validation where access to confidential data is restricted. These public datasets allow the research team to test anonymization techniques, evaluate

privacy-utility trade-offs, and demonstrate system functionality without exposing real confidential records.

The data collection process includes:

- Obtaining microdata from the Department of Census and Statistics.
- Reviewing dataset structure, column names, and metadata.
- Identifying direct identifiers, quasi-identifiers, sensitive attributes, and non-sensitive attributes.
- Using publicly available datasets for privacy-safe experiments and model testing.
- Preparing anonymized and synthetic versions of datasets for analysis.

2.2. Dataset Description:

Data source	Description	Resource	Size	Key attributes
1. Census Microdata (DCS Sri Lanka)	Primary dataset containing individual and household-level census information used for anonymization and analysis	Provided by Department of Census and Statistics (DCS)	Large-scale dataset (thousands to millions of records)	age, gender, occupation, region, household size, income
2. Publicly available / privacy-safe sample datasets	Sample datasets used for testing and basic validation of the system	Public datasets (online sources / open data platforms)	Small to medium datasets	age, gender, location, occupation (similar basic attributes for testing)

2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

	1	2	3	4
Data source 1	×	×	x	×
Data source 2	×	×	x	×

Data source 1, the census microdata provided by the Department of Census and Statistics, is relevant to all four research objectives because it contains real census-style attributes such as age, gender, district, occupation, education, income, household details, and other demographic information. These attributes are required for

identifier and quasi-identifier detection, anonymization technique selection, synthetic data generation, and re-identification risk evaluation.

Data source 2, the publicly available or privacy-safe sample datasets, is also relevant to all four objectives because it used for testing, experimentation, and validation without exposing confidential real records. It helps each component test its methods safely before applying them to the real census dataset. It also supports model testing, privacy-utility evaluation, synthetic data comparison, and risk analysis.

3. Methodology

3.1. Data Preprocessing:

Before applying the main analysis, the datasets were pre-processed to make them clean, consistent, and suitable for all four research components. The preprocessing stage included checking the dataset structure, standardizing column names, handling missing values, correcting data types, removing unnecessary indexing fields, and preparing attributes for privacy analysis.

Since the project focuses on census microdata anonymization, preprocessing was important for identifying direct identifiers, quasi-identifiers, sensitive attributes, and non-sensitive attributes. The cleaned data was then used for attribute classification, anonymization technique selection, synthetic data generation, and re-identification risk evaluation.

Objective

		Transformation technique							
		Data Cleaning	Missing Value Handling	Data Type Conversion	Data Encoding	Feature Engineering	Data Mapping	Data Normalization	Manual Review
Objective 1	Data source 1	×	×	×	×	×	×	×	Column names and sample values were

									reviewed to identify direct identifiers, quasi-identifiers, sensitive attributes, and non-sensitive attributes.
Objective 1	Data source 2	×	×	×	×	×	×	×	Public/sample datasets were checked to test identifier and quasi-identifier detection safely.
Objective 2	Data source 1	×	×	×	×	×	×	×	Attribute types and risk levels were reviewed to support suitable anonymization technique selection.
Objective 2	Data source 2	×	×	×	×	×	×	×	Sample datasets were used to test anonymization rules and technique recommendatio ns.
Objective 3	Data source 1	×	×	×	×	×	×	×	Direct identifiers were removed or marked before synthetic data generation, and sensitive attributes were reviewed.
Objective 3	Data source 2	×	×	×	×	×	×	×	Public/sample datasets were used to test differential privacy,

									permutation, and synthetic data generation methods.
Objective 4	Data source 1	×	×	×	×	×	×	×	Quasi-identifiers and sensitive attributes were reviewed to prepare data for re-identification risk analysis.
Objective 4	Data source 2	×	×	×	×	×	×	×	Sample datasets were used to test machine learning-based risk evaluation without exposing real records.

3.2. Scalability

The proposed framework is designed to be scalable across small, medium, and large microdata datasets. Since census and household survey datasets may contain thousands or millions of records, the system should support efficient data processing, automated attribute detection, and scalable anonymization workflows.

The dataset size is sufficient for analysis because it contains multiple attribute types such as demographic, geographic, household, income, education, and employment-related variables. These attributes allow the system to test different anonymization techniques and evaluate privacy risks under realistic conditions.

For larger datasets, the framework can be scaled by:

- Processing data in batches.
- Using optimized Python libraries such as Pandas, NumPy, and Scikit-learn.
- Applying sampling methods during early experimentation.
- Using distributed or cloud-based processing for very large datasets.
- Storing different dataset versions for reproducibility.
- Automating the anonymization and evaluation workflow through a Streamlit-based interface.

The framework is also scalable across different types of datasets because it does not depend only on one fixed dataset structure. Instead, it uses column-level analysis, metadata, feature extraction, and AI-based decision-making to adapt to new datasets.

3.3. Feature extraction

Objective 1

1. Key Features Evaluated

- Column-level characteristics such as data type (categorical/numerical) were analyzed to understand attribute behaviour
- Uniqueness percentage was evaluated to identify attributes with high re-identification risk
- Missing value percentage was examined to assess data quality and reliability
- Value distribution patterns (frequency of categories) were analyzed to detect imbalanced attributes
- Semantic meaning of column names (e.g., age, district, religion) was considered for attribute classification
- Combination potential of attributes (e.g., Age + Gender + District) was evaluated to identify possible quasi-identifiers

2. Feature Transformation

- Raw dataset columns were transformed into structured feature vectors for model input
- Categorical attributes were converted using encoding techniques
- Numerical features such as uniqueness %, missing % were normalized for consistency
- Column names and sample values were processed using text-based feature extraction for NLP understanding
- Combined statistical and semantic features were used to represent each column effectively
- Feature transformation enabled the model to differentiate between Direct Identifiers, QIs, Sensitive, and Other attributes

Objective 2

1. Key Features Evaluated

- Attribute category: direct identifier, quasi-identifier, sensitive, or non-sensitive.
- Data type: categorical, numerical, text, location, or date.
- Uniqueness percentage.
- Missing value percentage.
- Sensitivity score.
- Risk level of individual attributes and attribute combinations.
- Required data utility level.
- Suitability of anonymization techniques for each attribute type.

2. Feature Transformation

- Attribute-level statistics are converted into risk indicators.
- Sensitivity levels are mapped into low, medium, or high categories.
- Dataset characteristics are converted into decision rules.
- The AI agent uses these transformed features to recommend anonymization methods.
- Recommended techniques are stored with explanations for transparency.
- The selected techniques are passed into the anonymization pipeline.

Objective 3

1. Key Features Evaluated

- Attribute data type: categorical, numerical, ordinal, or text.
- Distribution of each column.
- Frequency of categorical values.
- Range, mean, standard deviation, and quartiles for numerical variables.
- Missing value patterns.
- Correlations between variables.
- Sensitivity level of each attribute.
- Quasi-identifier combinations that increase re-identification risk.

2. Feature Transformation

- Direct identifiers are removed before synthesis.
- Categorical distributions are converted into DP-protected marginal probabilities.
- Numerical values are transformed into bins or grouped ranges.
- Laplace noise is added to protect individual contributions.
- Permutation is applied to reduce direct record linkage.
- Synthetic records are generated using privacy-protected distributions.
- Output data is compared with original data to evaluate utility and privacy.

Objective 4

1. Key Features Evaluated
   - The similarity between anonymized and auxiliary records was evaluated using multiple feature comparison techniques such as age difference, gender matching, and location similarity.
   - Machine learning model prediction probabilities were used to estimate the likelihood of re-identification for each record.
   - Aggregated risk scores such as maximum and average prediction values were calculated to determine record-level vulnerability.
   - Feature importance analysis was performed to identify the most influential attributes contributing to re-identification risk.
   - The distribution of prediction scores was analyzed to understand how risk varies across different records.

2. Feature Transformation

   - Raw comparison features were transformed into numerical values suitable for machine learning models.
   - Similarity-based features such as age difference and categorical matches were normalized to maintain consistency.
   - Prediction probabilities obtained from trained models were aggregated to compute overall risk scores for each record.
   - Risk scores were converted into percentile-based values to ensure fair and balanced classification.
   - Records were categorized into different risk levels such as Critical, High, Medium, and Low based on score distribution.

4. Modelling and Results
   4.1. Key Insights:

Objective 1

- Quasi-identifiers are not risky individually, but attribute combinations significantly increase re-identification risk
- Attributes such as age, gender, and district frequently appear in high-risk combinations
- High uniqueness values directly correlate with increased disclosure risk
- Some attributes classified as non-sensitive can still contribute to risk when combined

- Rule-based methods are effective for direct identifiers, but machine learning is essential for QI detection
- Automated detection improves efficiency compared to manual selection used in traditional SDC tools

Objective 2

- Different attributes require different anonymization techniques.
- Direct identifiers should usually be removed or masked.
- Quasi-identifiers often require generalization, grouping, permutation, or k-anonymity.
- Sensitive numerical variables may require differential privacy or controlled noise addition.
- An AI-based recommendation approach can reduce manual decision-making and improve consistency.
- Privacy-utility balance is important because excessive anonymization can reduce dataset usefulness.

Objective 3

- Synthetic data generation helps reduce direct exposure of real individual records.
- Differential privacy provides mathematical privacy protection by adding controlled noise.
- Permutation helps preserve useful statistical patterns while breaking direct record-level relationships.
- Utility metrics such as TVD, KS statistic, missingness comparison, and distribution similarity help evaluate synthetic data quality.
- Privacy metrics such as KNN distance and exact match analysis help measure disclosure risk.
- A good synthetic dataset should be statistically similar to the real dataset but not too close to individual real records.

Objective 4

- The machine learning models achieved high performance in identifying matching records, with evaluation metrics such as:
    - ROC-AUC ≈ 0.99

- PR-AUC ≈ 0.98

- The final model was selected based on the highest PR-AUC value, as it provides better evaluation for imbalanced datasets where matching records are limited.
- The analysis revealed that a large proportion of records fall under Critical and High Risk categories, indicating that many anonymized records are still vulnerable to re-identification attacks.
- Feature importance analysis showed that attributes such as age, gender, and location significantly influence the risk of re-identification, as these attributes can uniquely identify individuals when combined.
- Overall, the results demonstrate that machine learning techniques can effectively simulate real-world attack scenarios and provide meaningful insights into privacy risks.

4.2. Challenges Faced During Data Analysis:

Objective 1

- Difficulty in clearly distinguishing between Quasi-Identifiers and Sensitive attributes
- Limited availability of labelled data for training the classification model
- Variations in dataset structure across different sections made analysis complex
- Some attributes had missing or inconsistent values, affecting reliability
- Identifying meaningful feature representations for columns required both statistical and semantic understanding
- Ensuring the model generalizes well to new unseen datasets was a key challenge

Objective 2

- Selecting the most suitable anonymization technique for each attribute was complex.
- Some attributes required more than one privacy-preserving method.
- Defining clear decision rules for the AI agent was challenging.
- Privacy risk and data utility often conflicted with each other.

- The system needed to provide explainable recommendations rather than black-box decisions.
- Different datasets may require different anonymization strategies.

Objective 3

- Maintaining a balance between privacy and utility was challenging.
- Adding too much noise reduced data usefulness.
- Adding too little noise increased privacy risk.
- Preserving correlations between variables was difficult during synthetic data generation.
- Handling mixed data types such as categorical and numerical variables required different methods.
- Evaluating whether synthetic data was safe and useful required multiple metrics.

Objective 4

- One of the main challenges was handling imbalanced data, as the number of matching record pairs was significantly lower than non-matching pairs. This required careful model evaluation using PR-AUC.
- Another challenge was converting pair-level predictions into record-level risk scores, which required aggregation techniques such as maximum and average scoring.
- Selecting appropriate thresholds for classifying risk levels was difficult, and therefore percentile-based methods were used to ensure balanced categorization.
- Some features had low correlation with risk, making feature selection and interpretation more complex.
- Ensuring that the model results were interpretable and understandable required additional analysis of feature importance.

5. References

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